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BanglaMedNER: A Gold Standard Medical Named Entity Recognition Corpus for Bangla Text

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***Abstract*—Medical Named Entity Recognition (MNER) has gained much attention recently and aims to recognize medical terms from text, such as proteins, diseases, symptoms, drugs, chemicals, and human anatomy. However, MNER for Bangla text has yet to progress significantly due to various challenges. One of these challenges is the need for a large and standardized corpus in this domain for Bangla. This work presents the annotated Bangla corpus for medical named entity recognition in standard IOB2 format consisting of more than 117000 tokens. The annotation process involved collecting texts from diverse Bangla health and drug-related domains. This corpus includes three distinct entity classes: Chemicals and Drugs (CD), Disease and Symptom (DS), and Anatomy (ANAT). Multiple machine-learning techniques were explored for medical named entity recognition, and the Bidirectional Long Short-Term Memory (BiLSTM) model, coupled with a Bidirectional Conditional Random Field (BI-CRF) layer, outperformed all the other models in our dataset with an outstanding F1 macro average score of 75%. We also employed class weights to account for the biases in our dataset.**

***Index Terms*—Medical Named Entity Recognition; Annotated Dataset; Machine Learning Techniques; SGD; Bidirectional LSTM-CRF.**

I. Introduction

Proper treatment is a primary need of human beings. In this digital era, we depend on various advanced technologies that produce medical text data daily to provide appropriate treatment. Medical text mining has received more attention recently as a result of the increasing number of reports, research articles, books, and other publications that are easily accessible in electronic format. Named Entity Recognition (NER) is performed to analyze and extract data from such texts belonging to the information extraction subfield [1]. The goal of Medical Named Entity Recognition (MNER) is the automatic and accurate extraction of medical named entities. It assists researchers in various situations, such as diagnosing diseases in a specific location, developing medications, and defining their side effects etc.

There are 228 million native speakers of Bangla worldwide. Thus, improving the linguistic abilities of languages like Bangla is essential [2]. Bangla Medical Named Entity Recognition (MNER) is an indispensable tool in the healthcare domain, as it assists healthcare professionals in comprehending patient data, conducting efficient analysis of medical records, monitoring social media platforms for medical issues, recommending specialized physicians, facilitating medical research endeavors, ensuring the safety of pharmaceutical products, and automating the processes of medical billing and coding. In Bangla-speaking countries, the utilization of this technology has been shown to improve efficiency, enhance patient care, and provide valuable medical insights. Nevertheless, tasks like medical named entity recognition have yet to see much progress in the current state of Bangla Natural Language Processing (NLP) because Bangla is more complicated in terms of usability and vocabulary [3] than high-resource languages like English, Chinese, etc.

A range of works have been done in medical named entity recognition in other languages like English, Chinese, and Spanish. These languages also have several corpora, including several medical entity classes such as drugs, chemical compounds, symptoms, genes, human body parts, proteins, etc. There are several difficulties to Bangla Medical Named Entity Recognition (MNER), including a shortage of annotated datasets, a challenging writing system, uncertainty in entity boundaries, syntactic and semantic complexity, uncommon terms, and a lack of standardization in named entities. These challenges make it difficult to develop reliable Medical NER models for Bangla. Many methods have been used in those languages to recognize entities. The rules-based technique [4] was also used, but it took too long to generate all the rules to identify medical entities. Several machine learning approaches, such as Maximum Entropy (ME) [5], Convolutional Neural Network (CNN), Recurrent Neural Network (RNN) [6], and BiLSTM CRF [7], give good accuracy. In recent work, transformer models [8] provide better accuracy. There is no standard publicly available Bangla corpus, and no noteworthy work has been done to recognize medical entities in the Bangla language. In our study, we developed a Bangla Medical Named Entity corpus with three classes and experimented with different methods to recognize medical named entities in the Bangla language.

The remaining sections are arranged as follows. Section II describes some previous studies related to this field. The building method includes the corpus data collection process, annotation process, and properties of our corpus in section III. We present our medical named entity recognition system in section IV. Experimental analysis is shown in Section V, and finally, Conclusions are drawn in Section VI.

II. Literature Review

Numerous medical corpora exist for English that can be used for various NLP applications. The GENIA corpus [9], GENETAG corpus [10], and CSIRO Adverse Drug Event Corpus (CADEC) [11] are now widely used. Several non-English MNER corpus [12]–[14] are also publicly accessible. Although Bangla Named Entity Recognition (NER) has received much interest [15], [16], Bangla Medical Named Entity Recognition has not received much attention, even though there is no publicly available medical named entity recognition dataset. A Bangla biomedical named entity corpus [17] is available, comprising only four classes. This corpus consists of a mere 818 sentences. Notably, the occurrence of medical entities within this corpus is relatively low, accounting for only 17% of the total tokens. Furthermore, it is worth mentioning that no baseline study has been conducted on this corpus thus far. Islam et al. [18] suggested an approach employing an attention-based BiLSTM-CRF model aimed at recommending specialized physicians to individual patients using NER. However, the paper notably lacks a comprehensive description of the training and testing procedures. Furthermore, the dataset comprises merely 4,000 words, with a limited frequency of medical terminology. This raises a challenge in training a robust model capable of effectively discerning medical nuances.

In our research, we created a Bangla medical named entity dataset to address the lack of resources in this critical area of NLP. The corpus was annotated with medical disease-related information. The dataset was manually tagged with the following IOB2 format using B-CD, I-CD, B-DS, I-DS, B-ANAT, and I-ANAT as classes. We evaluated various machine learning models for medical entity extraction using our Bangla gold MNER dataset. The best-performing model was the Bidirectional LSTM-CRF.

III. A Corpus for Bangla Medical Entity

We created our own MNER corpus for Bangla[1]. This work has three major segments: Collecting text data from websites by web scrapping, text pre-processing, and MNER dataset preparation.

We studied almost 64,000 drug-related articles [19], [20] from different sites and collected more than 300,000 sentences. The well-known open-source Python framework for web scraping was used to collect a lot of internet articles. A significant portion of the initially collected sentences exhibited repetition, rendering them redundant for the corpus's intended purpose. Furthermore, to enrich the corpus with varied linguistic expressions and contexts, additional sentences sourced from newspaper articles and social media posts were included, prioritizing quality and relevance in alignment with the corpus's objectives. MNER dataset preparation has two essential parts. We defined our MNER entity classes and then annotated each token according to the specified classes with the help of experts in this domain. We named our corpus as BanglaMedNER in the following.

A. Entity Classes

Three high-level entity classes are defined for the Bangla medical corpus:

Chemicals and Drugs (CD): This entity label denotes amino acid, peptide, protein, antibiotic, chemical, clinical drug, enzyme, hormone, pharmacological substance, and receptor. Some examples are - ইনসুলিন (Insulin), ভিটামিন বি (Vitamin B), নাপা এক্সটেন্ড (Napa Extend).

Disease and Symptom (DS): This class includes numerous diseases, anatomical abnormalities, acquired abnormalities, congenital abnormalities, syndromes, injuries, and mental dysfunction. Some of the examples of this category - জ্বর (Fever), হেপাটাইটিস বি (Hepatitis B).

Anatomy (ANAT): This entity name describes the anatomical structures, body parts, organs, organ components, tissues, cells, and cell components. Some examples include -মাথা (Head), হাত (Hand), ফুসফুস (lung), হৃদপিণ্ড (heart).

B. IOB2 Annotation Format

In order to apply language processing algorithms to Bangla MNER, we annotated each token by following the IOB2 [21] format. The IOB2 format is well-adopted, and all the annotated tokens are verified by a medical professional. The IOB2 (Inside-Outside-Beginning 2) format is a standard way to represent named entity recognition (NER) annotations in text data. In this format, each token in a sentence is tagged with its NER class, preceded by a prefix indicating its position with respect to the entity. The prefixes are:

B: Beginning of a medical named entity

I: Inside of a medical named entity

O: Outside of a medical named entity

An example of an IOB2 format annotated sentences is shown in Fig. 1.

C. Corpus Generation

The distribution of object types associated with IOB2 tags is shown in Table II. It illustrates the corpus's distribution of all types of entities. Since source articles include more information about numerous medicines and chemicals, CD is the most common entity type.

The entities falling within the ANAT category are those with the lowest observed frequency. It also indicates that non-medical entity mentions account for more than

[1]https://github.com/MuntakimJipu/BanglaMedNER-Dataset

Table I: Some of the examples of Medical Named Entity.

Class	Token
B-CD	ডেক্সন, ডেক্সামিথাসন, কর্টিসল, এইস, নিউরোট্রান্সমিটার, প্রিমেইস, এমডোক্যাল, হাইড্রোজেন, রুপাডে, ভেলোজেন, এমলোজেপ, টপিক্যাল, কুইবর্ণ, থিওফাইলিন, এন্টিস্পোজমোডিক
I-CD	গ্লুটামেট, ট্রাইফসফেটেজ, পটাশিয়াম-এডিনোসিন, স্টেরয়েড, ক্লিয়ারেন্স, আইভিঃ, ইউরিয়া, প্যারাসাইট
B-DS	আলসার, সেরেব্রোভাসকুলার, ঝিমুনী, ছত্রাক, মিউকোজাল, অরোফেরিনজিয়াল, ইসোফেগাইটিস, ক্রিপটোকক্কাস, বমি, রিভার, করোনভাইরাস
I-DS	ঝাপসা, তলপেটে, ব্যথা, ক্যান্ডিডিয়াসিসের, সংক্রমণের, ক্যানডিডিয়াসিস, বমি, ভাব, ডিসকমফোর্ট, এনজিওইডিমা, ব্লাইন্ড
B-ANAT	গ্যাস্ট্রিক, রক্তরসে, যকৃতের, পাকস্থলীর, লিভার, সেল-এসোসিয়েটেড, কোষের, ঘাড়, কিডনির, শিরা, ভ্যাজিনাল
I-ANAT	প্যারাইটাল, কোষের, ডিএনএর, সেল, ঝিল্লির, পেশী, শ্বসনতন্ত্রের, স্নায়ুতন্ত্রের

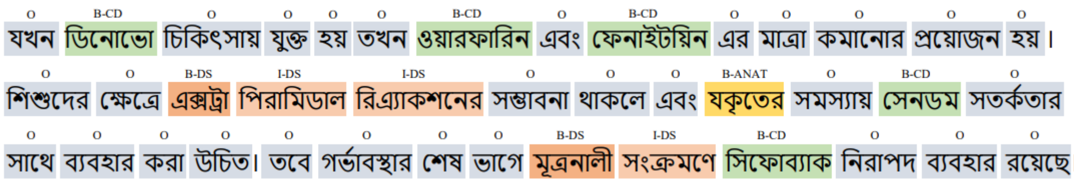

Figure 1: Sentence annotation with IOB2 format.

Table II: Frequency of the tokens in each class

Class	Frequency
B-CD	12163
I-CD	6104
B-DS	4207
I-DS	3301
B-ANAT	1273
I-ANAT	305
O	89394

75% of the textual content, as expected for any named entity corpus (Fig. 2). More statistical data about the BanglaMedNER corpus are shown in Table III.

Table III: Statistics of the BanglaMedNER corpus

Metric	Count
Number of Tokens	117152
Number of Sentence (Unique)	9000
Number of Words with Medical Entity	27353
Number of Non-entity Words	89494
Average sentence length (in words)	18.9

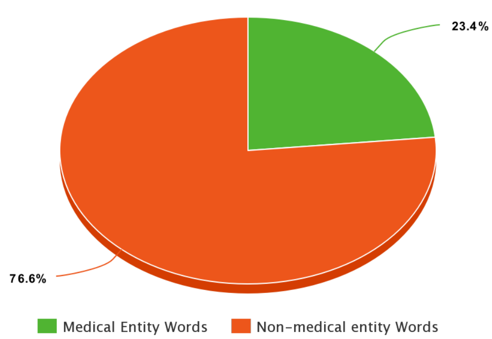

Figure 2: Distribution of medical vs non-medical entity in our corpus.

We divided our workflow into two broad steps (Fig. 3)- Bangla MNER dataset making, which was discussed in the previous section, and Bangla MNER System development.

D. Data Cleaning

Data cleaning is the process of recognizing and correcting errors, gaps, inconsistencies, and unconventional structures within datasets, which helps to make data more effective and insightful for analysis and meaningful interpretation. In our task, data contains unnecessary emojis and other special characters. We design our data for the next phase by cleaning those data from the dataset.

E. Preprocessing

After cleaning, data are converted into a list of tuples in the following format:

$$(Token_1, Tag_1), (Token_2, Tag_2), \ldots, (Token_n, Tag_n)$$

This process helps us with further feature extraction. In the next step, we pre-processed the text before we input the data into the model. We had to ensure that the text had the same length to feed the data to our model, which required the longest sentence in the sample. Smaller sentences are padded to make it longer.

F. Word to Index

The next step is to create a dictionary that gives each word in the corpus a distinct integer value. One hot encoding was used for tag encoding. This process saves a lot of memory. Then, we feed our machine learning models these integer vectors. We also created a reversed dictionary for decoding that links each index to a word.

IV. Experimental Analysis with Classification Algorithms

A. Classification Algorithms

1) Perceptron: The weighted total is calculated by multiplying input values by their corresponding weight values,

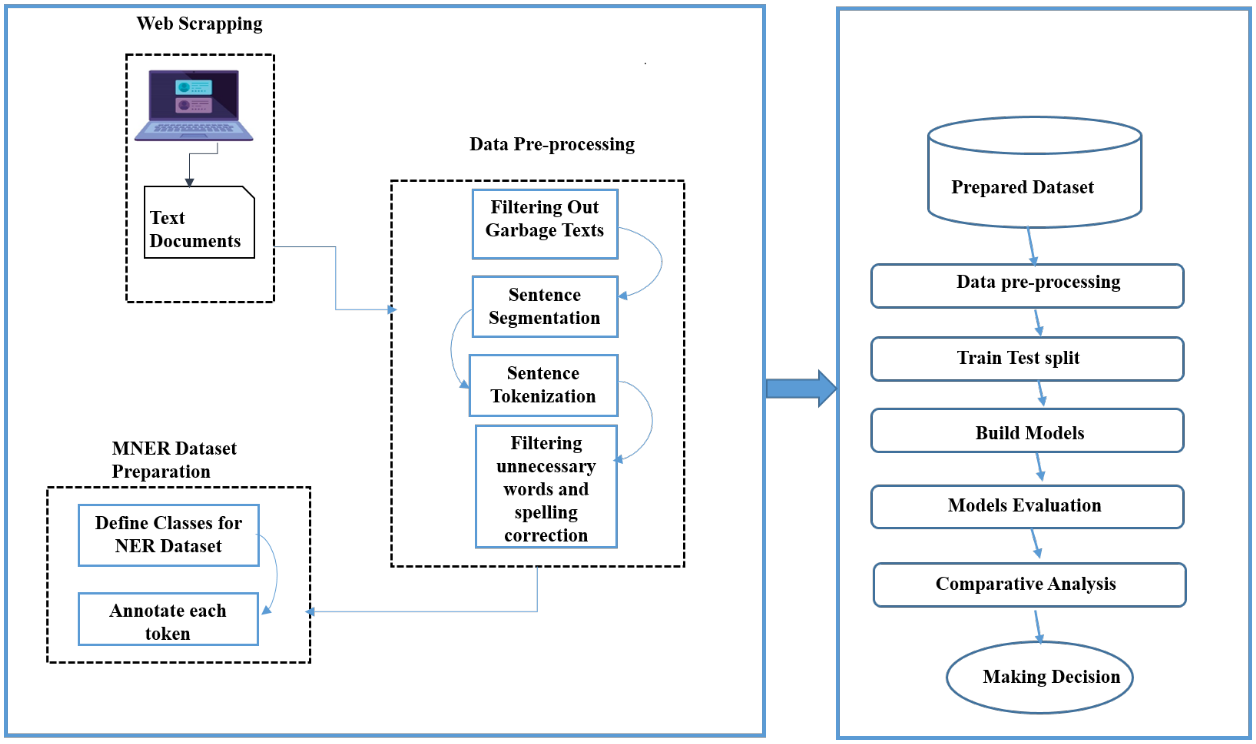

Figure 3: The proposed system in a nutshell.

adding the results, and adding the bias. An activation function is then applied to the weighted sum to provide output that can be binary or a continuous value. This process is used in a Perceptron [22], a single-layer neural network that makes predictions or classifies input data.

2) Passive Aggressive Classifier: The passive-aggressive classifier [23] is an online learning method for classification problems that allows weights to be updated with new information. It has a regularization parameter that balances the number of misclassifications and margin size. It evaluates whether an instance is accurately categorized after each iteration and modifies weights as required. Although it utilizes a different learning rule and does not call for a learning rate, it is similar to the Perceptron method. In contrast to Perceptron, which changes its model after each instance, it only does so when there is a mistake.

3) Stochastic Gradient Descent: In Stochastic Gradient Descent(SGD) [24], only a few random samples are chosen for each iteration instead of the entire dataset, making it more computationally efficient than GD as it performs iterations in batches of one. At each iteration, the SGD algorithm determines the gradient of the cost function for a single case. In SGD, the path traveled to the minima by the algorithm is typically noisier but less computationally expensive than in traditional gradient descent. The SGD algorithm can be represented as:

$$\begin{aligned}&\text{for } i \text{ in range}(m):\\&\quad \theta_j = \theta_j - \alpha(\hat{y}^i - y^i)x_j^i\end{aligned}$$

where m represents the number of samples, θ_j is the jth parameter, α is the learning rate, $\hat{y}^i$ is the predicted output, y^i is the actual output, and x_j^i is the jth feature of the ith sample.

4) Naive Bayes: Naive Bayes [25] is a simple probabilistic classifier that assumes features are unrelated to each other as long as the class variable is known. It uses feature vectors to calculate probabilities and identifies the class with the highest probability based on observed feature values. The class label of X is predicted as class y with the highest probability.

$$P(y, X) = P(y) \sum_{m=1}^{M} P(X_m|y) \tag{1}$$

5) Bidirectional LSTM-CRF: Bidirectional long-short-term memory (Bi-LSTM) is a neural network technique that includes sequence information in both directions. It's faster than one-directional LSTM and is useful for entity recognition problems. To improve accuracy, Bi-LSTM can be combined with a Bidirectional Conditional Random Field (BI-CRF) hidden layer [26], which provides better output values for nodes using the probabilistic method.

Table IV: Result(precision, recall, and F1 score) on test set of our MNER corpus. PAC represents a passive-aggressive classifier, and SGD represents stochastic gradient descent; balanced means class weight = balanced.

Model	**Accuracy**	**Precision**		**Recall**		**F1-score**	
		Macro	**Micro**	**Macro**	**Micro**	**Macro**	**Micro**
Perceptron	0.89	0.73	0.9	0.64	0.89	0.65	0.9
Perceptron (balanced)	0.89	0.66	0.90	0.67	0.89	0.61	0.89
PAC	**0.90**	0.70	0.90	0.68	0.90	**0.69**	0.90
PAC (balanced)	**0.91**	0.75	0.91	0.68	0.91	**0.71**	0.91
SGD	0.90	0.81	0.90	0.60	0.90	0.68	0.90
SGD (balanced)	0.90	0.62	0.91	0.75	0.90	0.66	0.91
Naive Bayes	0.88	0.63	0.89	0.58	0.88	0.60	0.88
Bidirectional LSTM-CRF	**0.98**	0.77	0.98	0.74	0.98	**0.75**	0.98

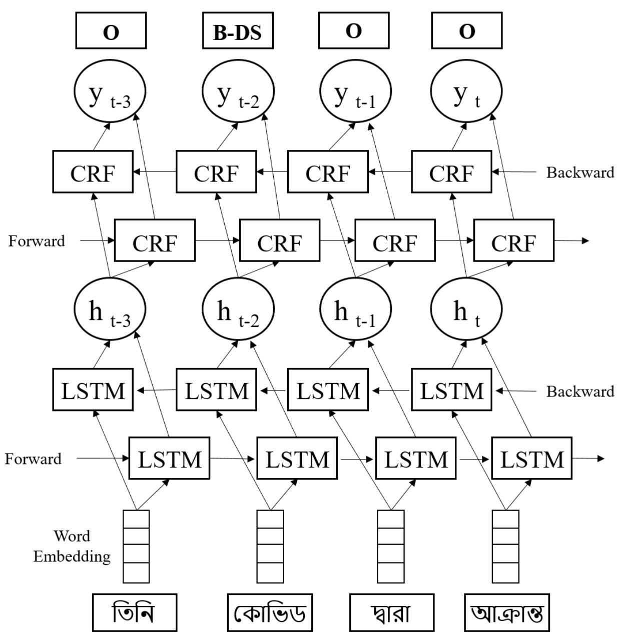

Figure 4: Bidirectional LSTM-CRF model data flow from input to output.

The linear chain of CRF distribution p(y|x) is given by (2).

$$P(y|x) = \frac{1}{Z(x)} \prod_{i=1}^{n} P(y_i, x_i) \tag{2}$$

Where Z(x) is the normalized function (3).

$$Z(x) = \sum_{c} \prod_{i=1}^{n} P(y_i, x_i) \tag{3}$$

To reduce bias or skewness in machine learning algorithms, we can assign weights to the majority and minority classes differently during training. This method addresses class imbalance and improves minority class predictions. Perception, Passive Aggressive Classifier, and Stochastic Gradient Descent models used this method. This method can improve classification accuracy by penalizing minority misclassification and changing class weights.

B. *Train-Test Split*

The split is random to provide a representative sample of text data in both training and test sets, with the proportion of data for training and testing depending on task requirements and available data. We used an 80-20 split with 80% of the data used for training and 20% for testing.

V. Results and Discussion

A. *Evaluation Metrics*

When evaluating Medical NER models, important metrics to consider are accuracy, precision, recall, and F1-score. However, it is crucial to consider bias when working with a biased dataset. In such cases, a balanced F1-score that considers both precision and recall provides a better indication of the model's overall effectiveness. The F1-macro score is a more appropriate metric for evaluating NER models on biased datasets since it offers a more balanced assessment of the model's performance across all classes and can detect biases that may not be easily apparent using the F1-micro score.

B. *Results*

Table IV shows the different performance metrics of each model of our study. Among them, the Bidirectional LSTM-CRF model outperformed all the other models with a macro F1 score of 75% and an accuracy of 98%. The second-best model was the passive-aggressive classifier (PAC). For the class as unbalanced macro F1 score was 69% and accuracy 90%. For the class as balanced macro F1 score was 71% and accuracy 91%.

Due to its capacity to recognize long-range dependencies, complicated label transitions, and contextual information, Bidirectional LSTM-CRF architecture performed well in our dataset. In addition to taking into account dependencies between labels, Bidirectional CRF layers also helped to ensure label consistency and coherence within a series of labels, both of which are vital in our work. Using class weight, we obtained more accurate and balanced results. Our models automatically assigned the class weights inversely proportional to their respective frequencies. It improved model generalization, particularly for less frequent classes, and boosted our accuracy by 5% to 7%.

VI. Conclusions

In this work, we have introduced a corpus of annotated medical named entities in Bangla, extracted from various health-related articles. This corpus represents the largest annotated medical named entity dataset in Bangla, formatted in the standard IOB2 style. The paper also provides detailed annotation guidelines and processes. This corpus contains over 117,000 tokens and comprehensive statistics for three critical medical entity classes. The research explores multiple models, including the Perceptron model, Passive Aggressive Classifier, Stochastic Gradient Descent, Naïve Bayes, and the deep learning model-Bidirectional LSTM (Bi-LSTM), enriched with a Bidirectional CRF layer for Bangla medical entity extraction. A comparative analysis of these machine learning techniques was conducted, and class weights were used to reduce the impact of biased datasets. The future plan incorporates all medical terms from the medical corpus, extending beyond disease-related terms to enhance the dataset's inclusivity and comprehensiveness.

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